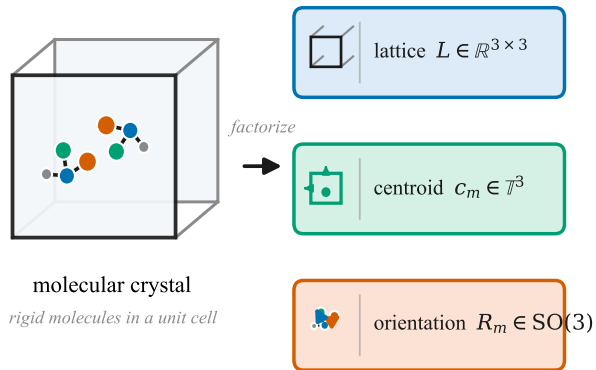


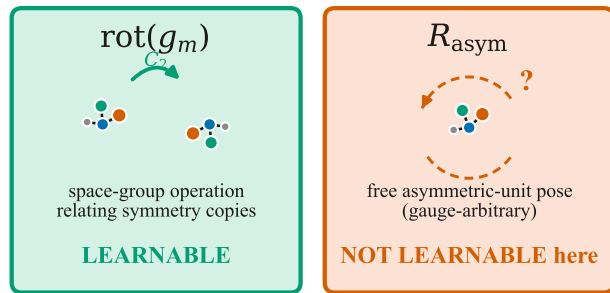
(a) Rigid-body factorization



$$\text{cart}_{m,a} = c_m L + R_m \text{local}_{m,a}$$

(b) Orientation target decomposes

$$R_m = \text{rot}(g_m) \cdot R_{\text{asym}}$$



evidence: +27% held-out loss,
13.7% packings reconstructed

evidence: 0% beyond
the predict-zero floor